

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE — GRADE 10 | SUB-STRAND 1.5: RESPIRATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation is YOUR chance to show everything you have learned across all 8 lessons on respiration. You are answering the big driving question that started with a puzzling thermos of uji in Kisumu. Your explanation must connect the science you have learned to that real phenomenon AND to broader uses of respiration in your community and economy. Read each section prompt carefully, then write your response. Use scientific vocabulary correctly, support every claim with evidence, and make clear links between the molecular processes and their real-world impacts. Write in full sentences and paragraphs. Your teacher will score your work using the rubric at the bottom of this document.

Section 1: Explain the Phenomenon — What Is Actually Happening in the Uji?

Go back to where this unit began. A student in Kisumu notices bubbles forming and an increasingly sour smell inside her sealed thermos of fermented uji. Using what you now know about respiration, explain:

- What living organisms are present in the fermented uji, and why they are NOT found in fresh uji (uji wa kawaida).
- Which type of respiration these organisms are carrying out, and why — name the process precisely.
- What molecules are being

The fermented uji contains yeast and lactic acid bacteria — microscopic living organisms that are NOT present in freshly cooked uji wa kawaida because the heat of cooking kills microorganisms and the porridge has not yet been exposed to the microbes that cause fermentation. Over time, these organisms colonise the uji and begin breaking down the sugars (mainly glucose, $C_6H_{12}O_6$) present in the maize for energy.

Because the thermos is sealed, oxygen levels inside quickly drop as the organisms use up whatever oxygen was present. With no oxygen available, aerobic respiration cannot continue. Instead, the yeast switch to anaerobic respiration — specifically alcoholic fermentation — producing ethanol and carbon dioxide gas (CO_2). The bacteria carry out lactic acid fermentation, producing lactic acid as a waste product.

The bubbles rising in the flask are carbon dioxide gas produced by yeast during alcoholic fermentation. Even though the thermos is sealed and no gas can escape, the CO_2 dissolves in the liquid and then comes out of solution when the flask is opened and pressure drops — exactly like opening a bottle of soda. The sour taste comes from the accumulation of lactic acid (and organic acids) produced by the bacteria, which lowers the pH of the uji, making it increasingly acidic over time.

This is why fresh uji does not change the same way: it lacks the microorganisms needed to carry out fermentation, so no CO_2 is produced and no acids accumulate.

broken down (the substrate), and what new molecules are being produced. – Where the bubbles come from, and what causes the sour taste. – Why the thermos being sealed matters for which type of respiration occurs.	
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Section 2: Compare Aerobic and Anaerobic Respiration at the Molecular Level	
Now zoom in to the chemistry. Compare aerobic and anaerobic respiration by addressing ALL of the following: – Write the word equation AND the balanced chemical equation for aerobic respiration. – Write the word equations for BOTH forms of anaerobic respiration (alcoholic fermentation and lactic acid fermentation). – Identify where in the cell each process takes place. – Explain why aerobic respiration releases far more ATP (energy) than anaerobic respiration. – Describe what happens to ATP once it is made — how does a cell actually USE that energy?	Aerobic respiration is the complete breakdown of glucose using oxygen to release the maximum amount of energy stored in the molecule. Word equation: Glucose + Oxygen → Carbon dioxide + Water + Energy (ATP) Balanced chemical equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \sim 38 \text{ ATP}$ Aerobic respiration happens in two main locations inside the cell: glycolysis occurs in the cytoplasm, and the Krebs cycle and electron transport chain occur in the mitochondria — which is why mitochondria are often called the 'powerhouse of the cell.' Anaerobic respiration also begins with glycolysis in the cytoplasm but does NOT use the mitochondria, because oxygen is absent. – Alcoholic fermentation (in yeast): Glucose → Ethanol + Carbon dioxide + Energy (2 ATP) – Lactic acid fermentation (in bacteria and muscle cells): Glucose → Lactic acid + Energy (2 ATP) Aerobic respiration yields approximately 38 ATP per glucose molecule, while anaerobic respiration yields only 2 ATP per glucose molecule. This huge difference exists because aerobic respiration fully oxidises glucose through the Krebs cycle and the electron transport chain, stripping all available hydrogen atoms and using oxygen as the final electron acceptor. Anaerobic respiration stops after glycolysis; the remaining energy is still locked inside the ethanol or lactic acid molecules and is never harvested. Once ATP is produced, the cell uses it as a universal energy currency. ATP releases energy by breaking the bond between its second and third phosphate groups ($ATP \rightarrow ADP + P_i + \text{energy}$). This released energy powers muscle contraction, active transport of ions across membranes, protein synthesis, cell division, and many other life processes.

Section 3: Respiration in the Human Body — From Resting to Running	
Think about what happens inside YOUR body during physical activity. Answer the following:	During rest, your muscle cells have a steady, sufficient supply of oxygen delivered by the blood. They therefore carry out aerobic respiration, completely breaking down glucose to produce approximately 38 ATP per glucose molecule. This is more than enough to power the low-level contractions needed to maintain posture and organ function.

- During a normal resting state, which type of respiration do your muscle cells use, and why is this sufficient? - Imagine you are a runner competing in the Eldoret Athletics Championships. Explain what happens to your muscle cells when you sprint at maximum speed and oxygen delivery cannot keep up with demand. - What molecule accumulates in your muscles during intense exercise, and what symptoms does this cause? - Explain what 'oxygen debt' means and what happens to the lactic acid after you stop running. - Describe how the circulatory and respiratory systems work together to support aerobic respiration during moderate exercise.	During a 100-metre sprint at the Eldoret Athletics Championships, your muscles demand energy far faster than your heart and lungs can deliver oxygen. When oxygen supply falls short, muscle cells switch to anaerobic respiration — specifically lactic acid fermentation. Glucose is broken down only as far as lactic acid, releasing just 2 ATP per glucose molecule but doing so very rapidly. This is what allows athletes to sustain maximum effort for a short time even without adequate oxygen. Lactic acid (lactate) accumulates in the muscle cells and bloodstream. This lowers the local pH, interfering with enzyme function and muscle contraction. You feel this as the characteristic burning sensation, muscle fatigue, and cramps that occur during or after intense effort. After you cross the finish line and stop sprinting, you continue breathing deeply and rapidly — this is 'oxygen debt' (also called excess post-exercise oxygen consumption, EPOC). The extra oxygen you take in is used by the liver to convert accumulated lactic acid back into glucose (through a pathway called gluconeogenesis) or to oxidise some lactic acid directly in the heart and other muscles to CO₂ and water. Your body is essentially 'paying back' the oxygen it borrowed during the sprint. During moderate exercise — such as a long-distance run — the circulatory and respiratory systems work in partnership. Your breathing rate increases so more oxygen enters the blood in the lungs and more CO₂ is expelled. Your heart beats faster and pumps more blood per minute (cardiac output increases), delivering oxygenated blood to working muscles more quickly. This teamwork keeps aerobic respiration supplied with oxygen and removes waste products, allowing sustained energy production.
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Section 4: Economic and Community Applications of Respiration in Kenya

The KICD key inquiry question asks: 'How is aerobic and anaerobic respiration important in a society's economy?' Answer this question in depth by discussing at least THREE distinct applications. For each application: - Name the specific product or process. - State which type of respiration is involved and which organism performs it.	Respiration — particularly anaerobic fermentation — underpins several industries and community practices that are economically vital across Kenya. 1. Fermented Food and Beverage Production Many traditional Kenyan foods rely on anaerobic fermentation by yeast and bacteria. Uji itself is fermented to improve its nutritional value, digestibility, and shelf life — a critical benefit for families in areas without reliable refrigeration. Similarly, traditional fermented drinks such as muratina (made from the muratina fruit by the Kikuyu community) and busaa (fermented from maize or millet) are produced through alcoholic fermentation by yeast: $\text{Glucose} \rightarrow \text{Ethanol} + \text{CO}_2 + 2 \text{ATP}$. Commercially, the Kenyan baking and brewing industries — including bread factories in Nairobi and the East African Breweries — depend on yeast fermentation to produce CO₂ for leavening bread and ethanol for beverages. These industries employ thousands of Kenyans and contribute significantly to GDP. 2. Biogas Production for Rural Energy
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– Explain the economic or community benefit for Kenyan people. You may draw on examples from food production, agriculture, medicine, environmental management, or industry.	In many farming communities in the Rift Valley and Western Kenya, livestock dung and crop waste are fed into biogas digesters. Anaerobic bacteria break down organic matter through anaerobic respiration, releasing methane (CH₄) and CO₂ as biogas. Families use this biogas for cooking and lighting, reducing dependence on expensive kerosene and firewood. This lowers household energy costs, reduces deforestation, and decreases respiratory diseases caused by indoor smoke. The spent slurry from digesters is also a rich organic fertiliser, benefiting crop yields. 3. Composting and Soil Fertility in Agriculture Aerobic respiration by decomposer microorganisms (fungi and bacteria) is the engine of composting. Farmers in Kericho and around Lake Victoria pile crop residues, kitchen waste, and animal manure into compost heaps. Decomposers aerobically respire — breaking down organic molecules using oxygen and releasing CO₂, water, and heat — and in doing so, transform complex organic matter into nutrient-rich humus. This compost improves soil structure, water retention, and fertility, reducing the need for costly synthetic fertilisers and increasing yields of staple crops such as maize and beans. 4. Medical and Biotechnology Applications Aerobic and anaerobic microbial respiration is harnessed in the production of antibiotics, vitamins, and other pharmaceuticals. For example, certain fungi and bacteria are grown in large fermentation tanks where they respire and, as metabolic by-products, produce compounds used as medicines. Closer to everyday life, the production of yoghurt and fermented milk products (such as mursik, the traditional fermented milk of the Kalenjin community) relies on lactic acid bacteria performing lactic acid fermentation. These products provide affordable nutrition and support the dairy economy in counties such as Bomet and Nakuru.
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Section 5: Returning to the Driving Question — Your Final Answer	
You have now explained the phenomenon, compared the chemistry, connected respiration to the human body, and explored economic applications. In this final section, write a coherent, well-structured answer to the driving question: 'How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?' Your answer should: – Synthesise ideas from ALL	At the start of this unit, respiration seemed like a simple word for breathing. But the bubbling, souring uji in a sealed thermos in Kisumu revealed something much deeper: respiration is a molecular process happening invisibly inside every living cell — in yeast, in bacteria, and in our own bodies — every second of every day. All living organisms need energy to survive, grow, and respond to their environment. That energy is locked inside the chemical bonds of food molecules, especially glucose. Respiration is the set of chemical reactions cells use to break those bonds and release the energy in a usable form — ATP. When oxygen is available, aerobic respiration extracts the maximum energy: about 38 ATP per glucose molecule, releasing CO₂ and water as harmless waste products. When oxygen is absent or insufficient — as it is inside a sealed thermos or in sprinting muscle cells — organisms switch to anaerobic respiration, sacrificing efficiency for speed, producing only 2 ATP per glucose but doing so rapidly, and releasing ethanol, CO₂, or lactic acid as by-products. These 'waste' by-products of anaerobic respiration turn out to be anything but wasteful to human communities. The CO₂ that yeast breathe out makes bread rise and uji ferment. The ethanol they produce is the basis of beverages and biofuels. The lactic acid from bacterial fermentation preserves and enriches foods like mursik and yoghurt. The methane from anaerobic decomposers lights homes in rural Kenya. And the heat from aerobic decomposers in compost heaps rebuilds the soils that feed millions of Kenyans. What changed my thinking most during this unit is understanding that 'invisible' does not mean 'unimportant.' The organisms transforming uji inside a sealed thermos are performing the same fundamental chemistry as the mitochondria in a Kenyan marathon runner's leg muscles at the moment of crossing the finish line in Eldoret. The scale is different, but the molecular logic is the same: break glucose, capture energy, keep living.

previous sections (do not just repeat them — connect them). – Explain why 'respiration' means something much bigger than just breathing. – Reflect on what surprised you or changed your thinking during this unit. – Describe one new question you still have about respiration that you would like to investigate further.	One question I still want to investigate: Could the fermentation happening naturally in uji be optimised — by controlling temperature or the type of grain used — to produce a more nutritious and consistent product that could support food security in communities around Lake Victoria? I think the answer lies at the intersection of microbiology, chemistry, and community needs — which is exactly what this unit has shown me science can do.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific Accuracy of Respiration Processes	Aerobic and both forms of anaerobic respiration are explained with correct word and chemical equations, correct ATP yields, and precise identification of cellular locations (cytoplasm and mitochondria). All molecular claims are accurate and consistently applied across sections.	Aerobic and anaerobic respiration are correctly described with mostly accurate equations and ATP yields. Minor errors in chemical equations or cellular locations do not undermine overall understanding.	Some correct ideas about respiration are present, but equations are missing or significantly incorrect, ATP yields are confused, or aerobic and anaerobic processes are conflated or mislabelled.
Explanation of the Anchoring Phenomenon (Uji)	The student provides a complete, accurate explanation of ALL observable changes in the uji (bubbles, sourness, no change in fresh uji), correctly identifying the organisms responsible, the type of respiration occurring, the substrates consumed, and the products formed. The sealed thermos condition is explicitly linked to the switch to anaerobic respiration.	Most observable changes are explained correctly and linked to the correct type of respiration and organisms. The role of the sealed thermos is mentioned. One minor aspect may be incomplete or slightly inaccurate.	The student identifies that microorganisms are involved and that fermentation is occurring, but the explanation of specific products, the reason for anaerobic conditions, or the distinction between fresh and fermented uji is incomplete or inaccurate.
Connection to the Human Body and Physical Activity	The student accurately explains aerobic respiration during rest, the switch to lactic acid fermentation during intense exercise, the cause and symptoms of lactic acid accumulation, oxygen debt, and the role of the circulatory and respiratory systems. All claims are connected to specific biological	The student correctly explains the switch between aerobic and anaerobic respiration during exercise and describes oxygen debt. The role of the circulatory or respiratory system is mentioned. One mechanism may lack precise detail.	The student shows basic awareness that exercise causes a change in respiration type, but lactic acid, oxygen debt, or the role of supporting organ systems is poorly explained or missing.

	mechanisms.		
Economic and Community Applications (Kenya Context)	Three or more distinct applications are discussed with specificity — each names the product or process, the type of respiration, the responsible organism, and a clear economic or community benefit in the Kenyan context. Kenyan examples (places, foods, communities) are used accurately and meaningfully throughout.	At least two applications are discussed with correct links to respiration type and Kenyan context. Benefits are stated clearly. A third application may be present but less fully developed.	One or two applications are mentioned but lack specificity about the type of respiration, the organism involved, or the Kenyan benefit. General statements are made without meaningful connection to local context.
Synthesis, Communication, and Scientific Thinking	The final section synthesises ideas from all previous sections coherently, redefines respiration beyond breathing with insight, reflects genuinely on a conceptual shift in understanding, and poses a specific, investigable new question rooted in Kenyan community needs. Writing is clear, well-structured, and uses scientific vocabulary accurately throughout.	The final section connects ideas from most sections and shows genuine reflection. Scientific vocabulary is mostly used correctly. A new question is posed, though it may be broad rather than specific and investigable.	The final section summarises earlier points without synthesising them. Reflection is superficial or absent. Scientific vocabulary is inconsistently used, and the new question is either missing or too vague to be investigable.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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